Handmade E-Commerce Sales Analysis Portfolio

Statistical EDA, Trends, & Top Markets Insights Using Python

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Project Overview

An analysis of the statistical and business aspects of handicraft e-commerce sales in 2011. The company is based in the UK. There are several processes, such as examining data distribution and outliers before discovering monthly sales trends focused on top markets and best-selling products using Python.

Questions to Answer

EDA Questions

1. How are the data distributed for unit price, quantity, and sales?
2. Are there any outliers in the unit price, quantity, and sales data? If so, show the middle data along with the outliers.
3. What is the percentage of outliers in the unit price, quantity, and sales data?
4. Are there any correlations between unit price, quantity, and sales data?
5. How strong is the correlation between unit price, quantity, and sales data?

Business Questions

1. How much were the e-commerce sales in 2011?
2. What are the total sales, average sales, and number of items sold on the e-commerce platform in 2011? Show 5 countries that dominate the market.
3. What are the monthly sales figures of the five countries with the highest sales?
4. What kind of items became the best-sellers in 2011? (limit to 5)
5. What kind of products sell the most in the top five countries with the most sales? (limit to 5)

Dataset Overview

This is the original dataset imported from ecommerce.csv using Pandas library in Python

```
# import dataset & check the 5 first data
df = pd.read_csv('ecommerce.csv')
df.head()
```

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country
0	554697	21166	COOK WITH WINE METAL SIGN	1	5/25/2011 17:31	2.08	14584	United Kingdom
1	561038	82482	WOODEN PICTURE FRAME WHITE FINISH	2	7/24/2011 11:58	2.55	17114	United Kingdom
2	560552	23192	BUNDLE OF 3 ALPHABET EXERCISE BOOKS	1	7/19/2011 12:54	1.65	15311	United Kingdom
3	559884	85183B	CHARLIE & LOLA WASTEPAPER BIN FLORA	12	7/13/2011 11:34	1.25	16843	United Kingdom
4	544450	21789	KIDS RAIN MAC PINK	3	2/20/2011 12:23	0.85	17811	United Kingdom

This is the dataframe after adding new column 'Sales'

```
df['Sales'] = df['Quantity'] * df['UnitPrice']
df.head()
```

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country	Sales
0	554697	21166	COOK WITH WINE METAL SIGN	1	5/25/2011 17:31	2.08	14584	United Kingdom	2.08
1	561038	82482	WOODEN PICTURE FRAME WHITE FINISH	2	7/24/2011 11:58	2.55	17114	United Kingdom	5.10
2	560552	23192	BUNDLE OF 3 ALPHABET EXERCISE BOOKS	1	7/19/2011 12:54	1.65	15311	United Kingdom	1.65
3	559884	85183B	CHARLIE & LOLA WASTEPAPER BIN FLORA	12	7/13/2011 11:34	1.25	16843	United Kingdom	15.00
4	544450	21789	KIDS RAIN MAC PINK	3	2/20/2011 12:23	0.85	17811	United Kingdom	2.55

Dataset Overview

```
# check data info
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4870 entries, 0 to 4869
Data columns (total 9 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   InvoiceNo    4870 non-null   int64
 1   StockCode    4870 non-null   object
 2   Description  4870 non-null   object
 3   Quantity     4870 non-null   int64
 4   InvoiceDate  4870 non-null   object
 5   UnitPrice    4870 non-null   float64
 6   CustomerID   4870 non-null   int64
 7   Country      4870 non-null   object
 8   Sales        4870 non-null   float64
dtypes: float64(2), int64(3), object(4)
memory usage: 342.6+ KB
```

```
df.info()

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Data columns (total 9 columns):
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 0   InvoiceNo    4870 non-null   int64
 1   StockCode    4870 non-null   object
 2   Description  4870 non-null   object
 3   Quantity     4870 non-null   int64
 4   InvoiceDate  4870 non-null   datetime64[ns]
 5   UnitPrice    4870 non-null   float64
 6   CustomerID   4870 non-null   int64
 7   Country      4870 non-null   object
 8   Sales        4870 non-null   float64
dtypes: datetime64[ns](1), float64(2), int64(3), object(3)
memory usage: 342.6+ KB
```
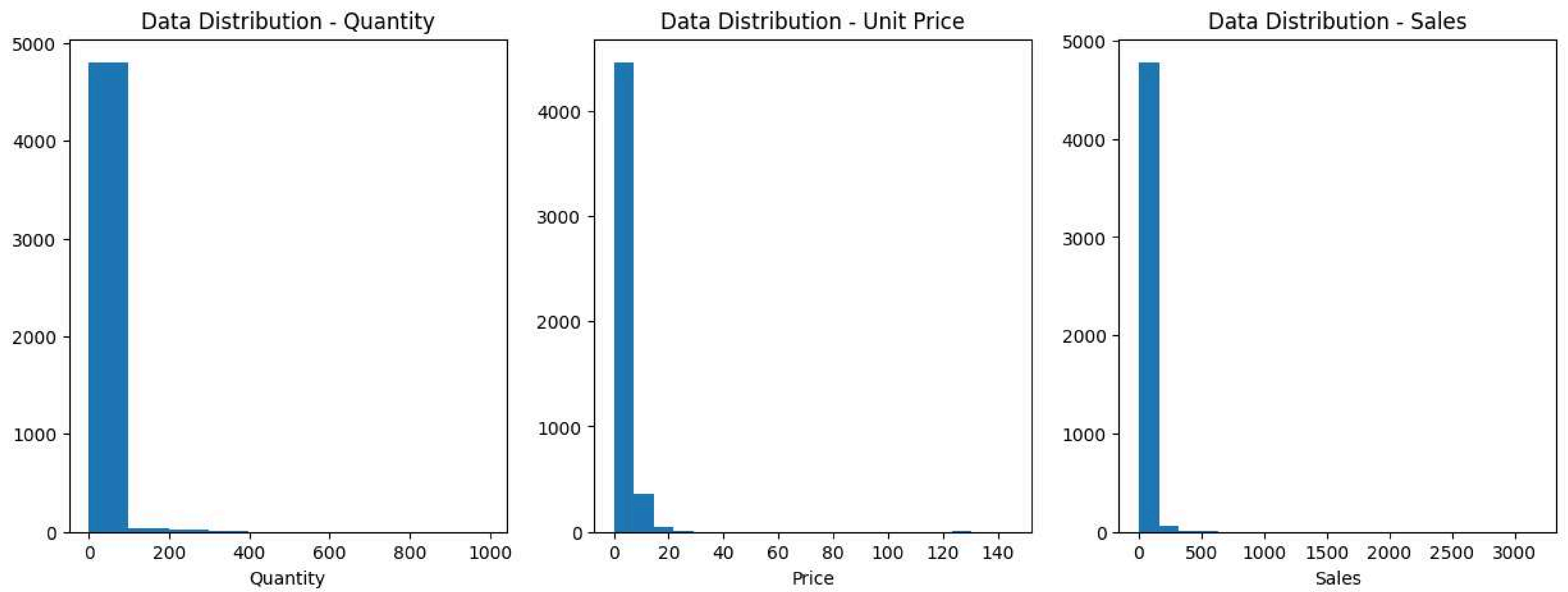
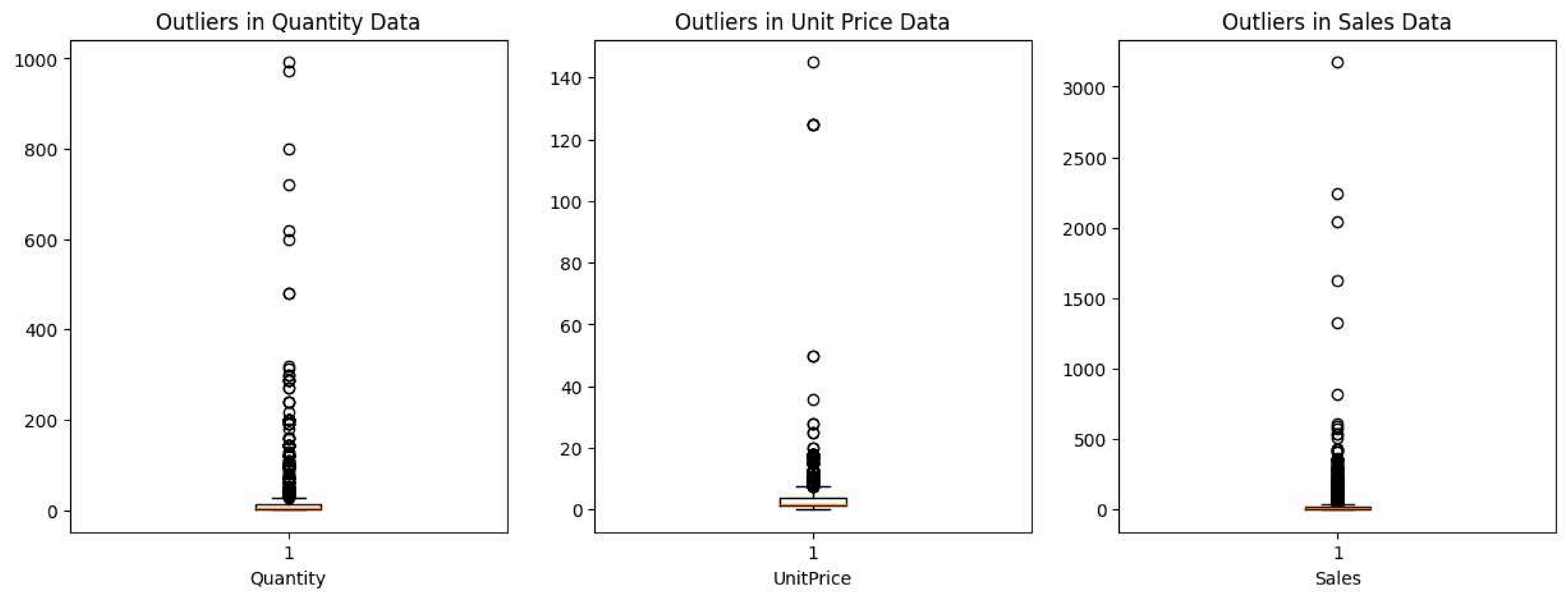

The data frame consists of 4870 rows and 9 columns. There are no empty data nor duplicated data, but the data type in InvoiceDate was incorrect. Therefore, it needs to be transformed into datetime.

Data Distribution Analysis



The histogram illustrated that the data from unit price, quantity, and sales tend to be right-skewed. This means that the data is centered on the low purchase with cheaper price yet the frequency of the transactions are quite often.

Outliers in Data



The boxplot exposed outliers existed in unit price, quantity, and sales data.

```python
# Check the outlier of unit price
Q1 = df['UnitPrice'].quantile(0.25)
Q3 = df['UnitPrice'].quantile(0.75)
range = df['UnitPrice'].max() - df['UnitPrice'].min()

IQR = Q3 - Q1

lower_line = Q1 - (IQR * 1.5)
upper_line = Q3 + (IQR * 1.5)

print('Mean vs Median Comparison (Middle Data):')
print('- Mean:', df['UnitPrice'].mean())
print('- Median:', df['UnitPrice'].median())
print('')
print('Data Distribution of Unit Price:')
print('- Std Price:', df['UnitPrice'].std())
print(f'- IQR Price: {IQR}')
print(f'- Lower Line: {lower_line}')
print(f'- Upper Line: {upper_line}')
print(f'- Range Price: {range}')
```

```
Mean vs Median Comparison (Middle Data):
- Mean: 2.9370574948665302
- Median: 1.95

Data Distribution of Unit Price:
- Std Price: 4.744514674613008
- IQR Price: 2.5
- Lower Line: -2.5
- Upper Line: 7.5
- Range Price: 144.96
```

```python
# Check the outlier of quantity
Q1 = df['Quantity'].quantile(0.25)
Q3 = df['Quantity'].quantile(0.75)
range = df['Quantity'].max() - df['Quantity'].min()

IQR = Q3 - Q1

lower_line = Q1 - (IQR * 1.5)
upper_line = Q3 + (IQR * 1.5)

print('Mean vs Median Comparison (Middle Data):')
print('- Mean:', df['Quantity'].mean())
print('- Median:', df['Quantity'].median())
print('')
print('Data Distribution of Product Quantities:')
print('- Std Quantity:', df['Quantity'].std())
print(f'- IQR Quantity: {IQR}')
print(f'- Lower Line: {lower_line}')
print(f'- Upper Line: {upper_line}')
print(f'- Range Quantity: {range}')
```

```
Mean vs Median Comparison (Middle Data):
- Mean: 12.945790554414785
- Median: 5.0

Data Distribution of Product Quantities:
- Std Quantity: 38.14598782418509
- IQR Quantity: 10.0
- Lower Line: -13.0
- Upper Line: 27.0
- Range Quantity: 991
```

```python
# Check the outlier of sales
Q1 = df['Sales'].quantile(0.25)
Q3 = df['Sales'].quantile(0.75)
range = df['Sales'].max() - df['Sales'].min()

IQR = Q3 - Q1

lower_line = Q1 - (IQR * 1.5)
upper_line = Q3 + (IQR * 1.5)

print('Mean vs Median Comparison (Middle Data):')
print('- Mean:', df['Sales'].mean())
print('- Median:', df['Sales'].median())
print('')
print('Data Distribution of Items Sold:')
print('- Std Sales:', df['Sales'].std())
print(f'- IQR Sales: {IQR}')
print(f'- Lower Line: {lower_line}')
print(f'- Upper Line: {upper_line}')
print(f'- Range Sales: {range}')
```

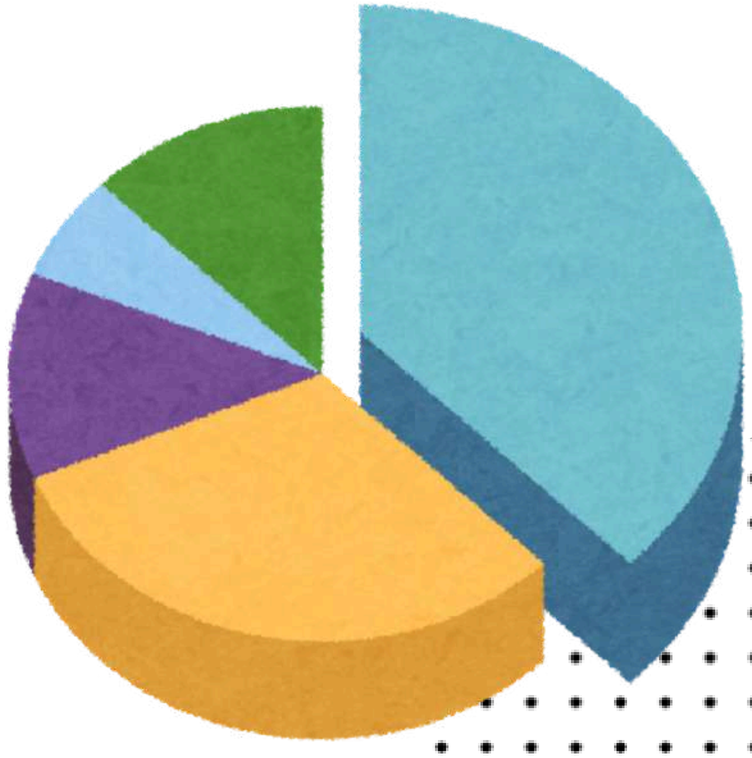
```
Mean vs Median Comparison (Middle Data):
- Mean: 22.952753593429158
- Median: 11.399999999999999

Data Distribution of Items Sold:
- Std Sales: 81.4088597754778
- IQR Sales: 15.549999999999997
- Lower Line: -19.074999999999996
- Upper Line: 43.12499999999999
- Range Sales: 3174.28
```

IQR was used to measure the midspread of the data and set boundaries.
All three data have their standard deviation surpassing either lower or upper lines, meaning that there are some data exceeding the normal boundaries.

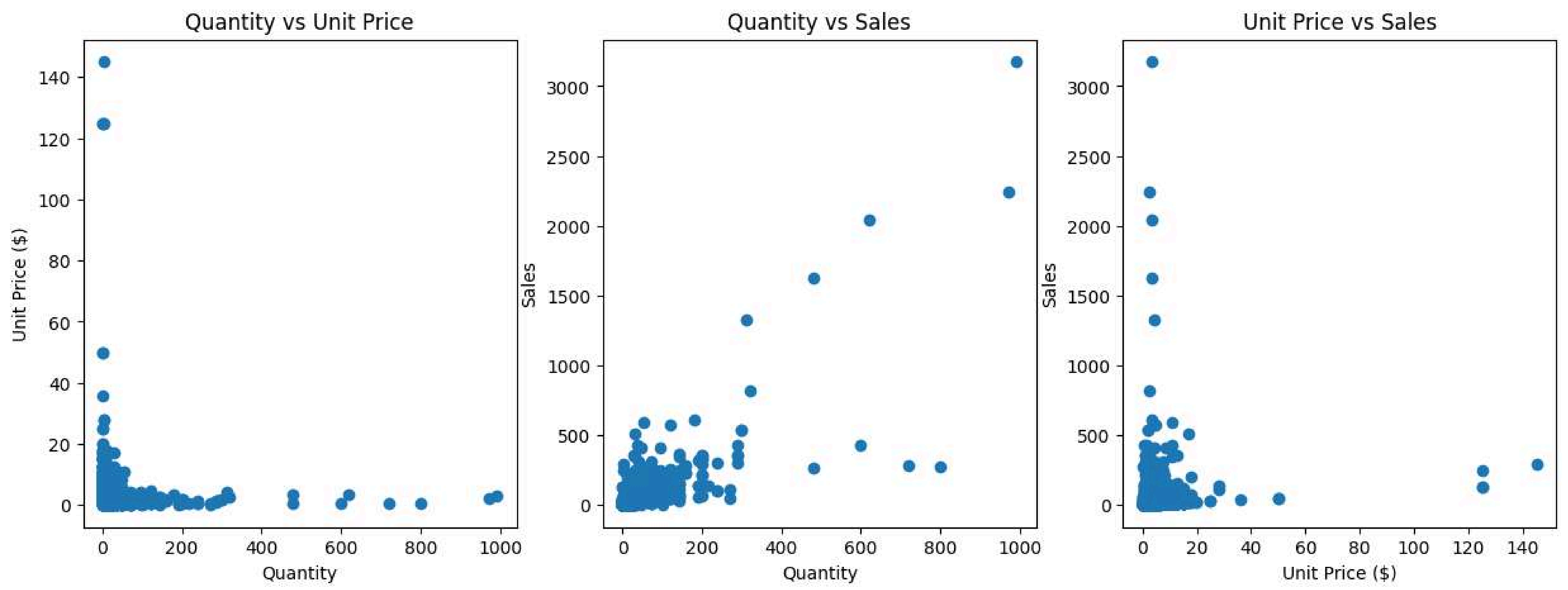
```python
price_percent = (len(price_outliers) / len(df)) * 100
qty_percent = (len(qty_outliers) / len(df)) * 100
sales_percent = (len(sales_outliers) / len(df)) * 100

print(f'The percentage of outliers in unit price: {price_percent:.2f}')
print(f'The percentage of outliers in product quantities: {qty_percent:.2f}')
print(f'The percentage of outliers in sales: {sales_percent:.2f}')
```

```
The percentage of outliers in unit price: 8.34
The percentage of outliers in product quantities: 6.74
The percentage of outliers in sales: 8.28
```

Overall, outliers in the unit price, quantity, and sales data do not exceed 10% of the total data. The outliers are kept because the amounts are still tolerable and will not make the data less accurate.



Correlation Analysis



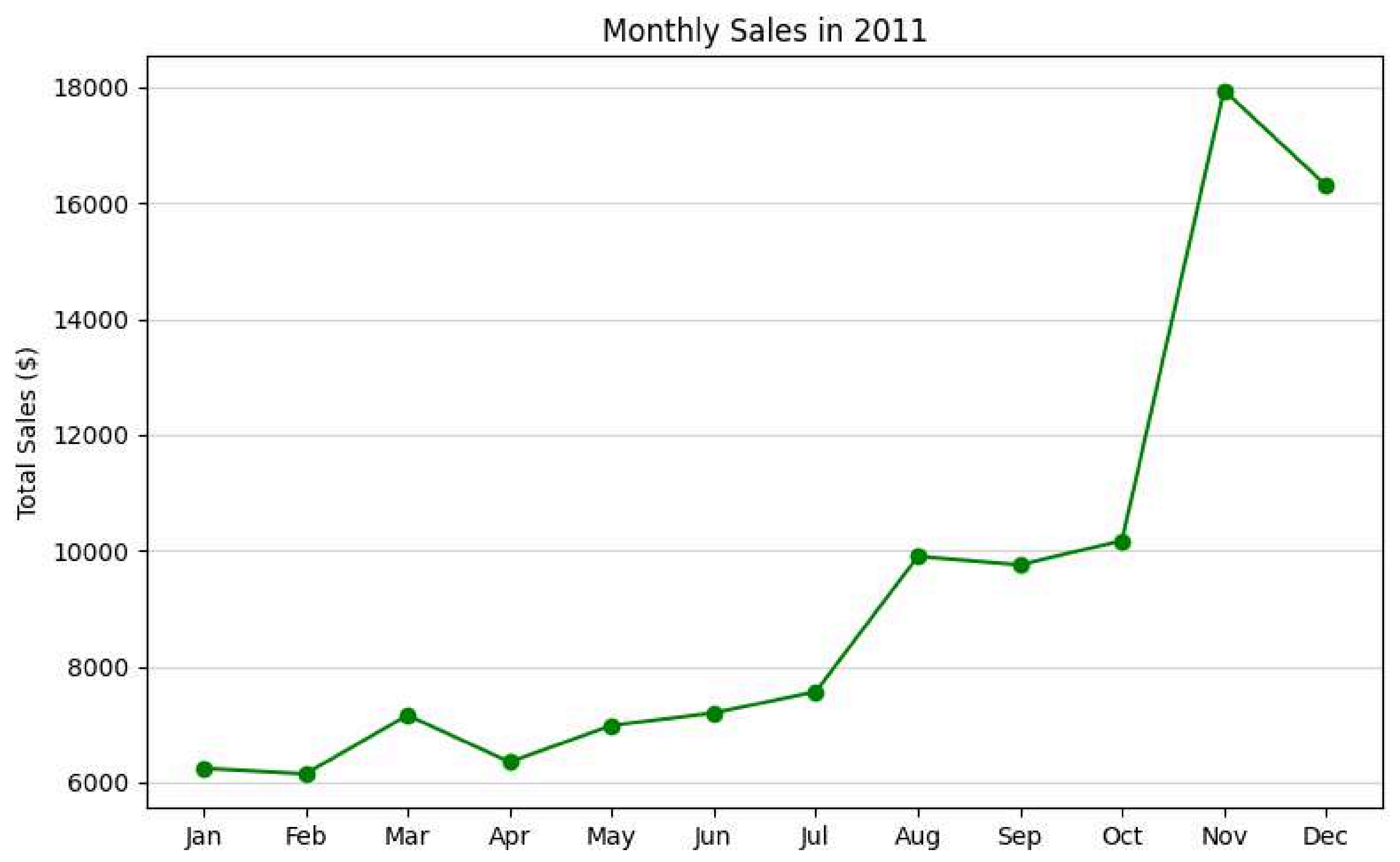
The scatterplots between Unit Price, Quantity, and Sales show us that:

1. There is a positive correlation between Quantity and Sales. The more items are sold, the higher number of sales.
2. There are no correlation between Quantity vs Unit Price and Unit Price and Sales.

Based on patterns from the heatmap graph, it can be concluded:

1. There is a strong positive correlation between Sales and Quantity.
2. There is nearly no correlation between Unit Price and Quantity.
3. There is nearly no correlation between Unit Price and Sales.

Monthly Sales Trend Analysis



Generally, in 2011, the company showed very significant growth.
From January to July, sales were relatively stable, but in August, transactions skyrocketed from 10,000 USD to nearly 18,000 USD in November, and then decreased slightly to almost 16,000 USD in December. Nevertheless, it was still considerably high.
The increasing numbers in sales might be caused by summer holidays and other events such as Halloween and Christmas, where giving presents is the norm.

Top 5 Countries by Sales

```
country_table = df.groupby(df['Country']).agg(
    Total_Sales =('Sales','sum'),
    Average_Sales =('Sales','mean'),
    Total_Products =('Quantity','sum')
).sort_values(by= 'Total_Sales', ascending=False)
country_table
```

1 to 25 of 31 entries Filter

Country	Total_Sales	Average_Sales	Total_Products
United Kingdom	93011.36	21.4016014726185	51694
Netherlands	3622.33	139.32038461538463	2473
Germany	3363.77	27.7997520661157	1742
EIRE	3156.78	35.07533333333333	1588
France	2320.55	26.07359550561798	1299
Sweden	988.74	82.395	1008
Australia	937.0	72.07692307692308	828
Switzerland	902.52	32.23285714285714	500
Portugal	562.9	35.18125	267
Finland	432.57	61.79571428571428	111
Belgium	418.96	19.950476190476188	242
Spain	405.19	16.2076	202
Japan	298.6	59.720000000000006	220
Norway	225.62	16.115714285714287	262
Greece	173.0	24.714285714285715	60
Austria	163.36	23.337142857142858	60
Cyprus	148.25	24.708333333333332	83
Channel Islands	137.39999999999998	17.174999999999997	106
Italy	95.85	19.169999999999998	39
Singapore	80.0	26.666666666666668	56
Denmark	51.18	12.795	59
Lebanon	47.400000000000006	47.400000000000006	12

```
# countries with top 5 sales
country_table.head(5)
```

Country	Total_Sales	Average_Sales	Total_Products
United Kingdom	93011.36	21.401601	51694
Netherlands	3622.33	139.320385	2473
Germany	3363.77	27.799752	1742
EIRE	3156.78	35.075333	1588
France	2320.55	26.073596	1299

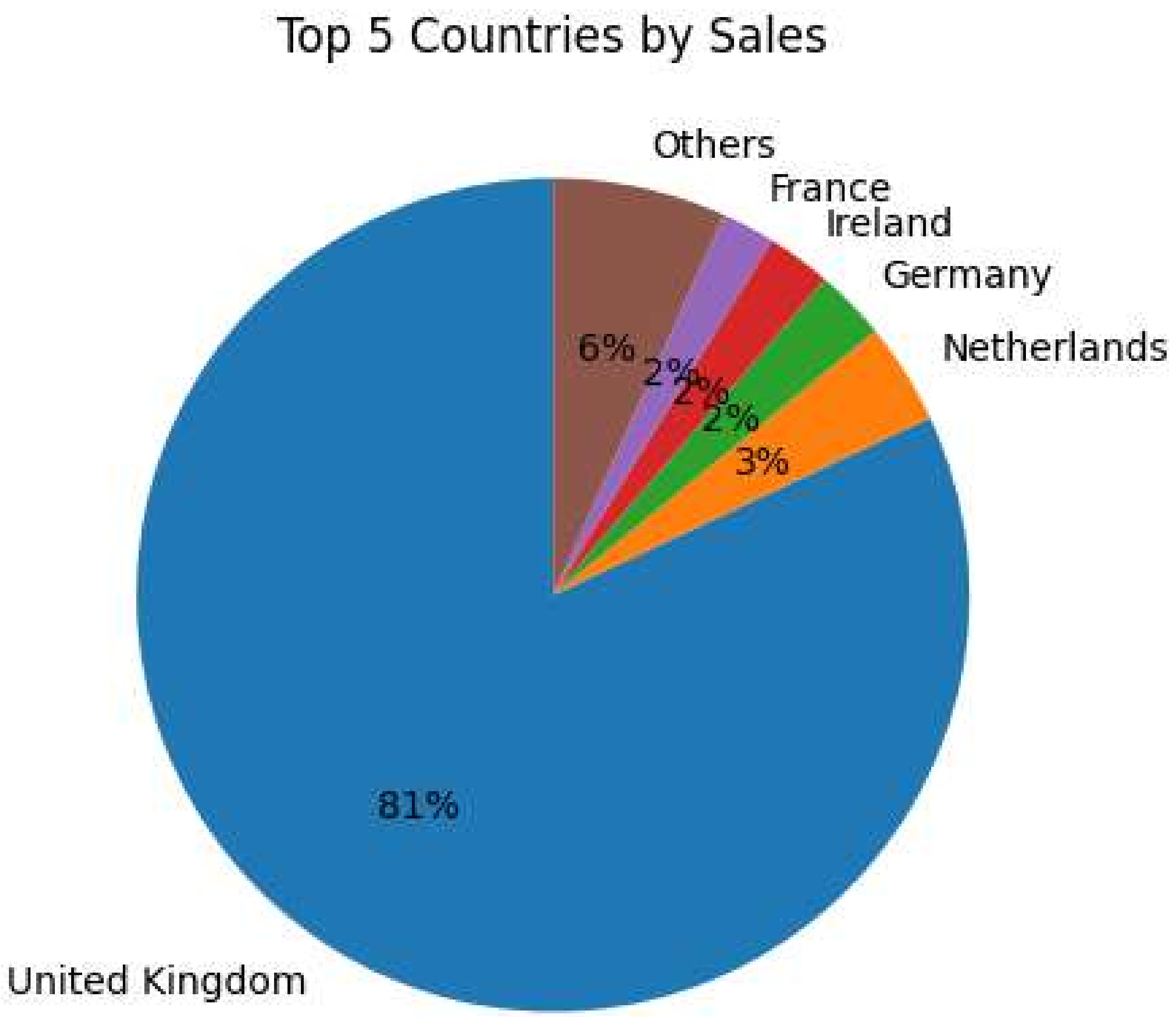
The dataset is filtered to show the total sales, average sales, and total items sold in every countries. Next, the data is only focused on the top 5 countries with the biggest sales.
The name of EIRE is also replaced to Ireland for famliarity purpose.

Top 5 Countries by Sales

The pie chart illustrates the distribution of the top 5 markets and the other markets. UK dominates 81% of the company's revenue, in contrast to the other countries combined.

There are 3 possibilities why the market is still centered in the UK:

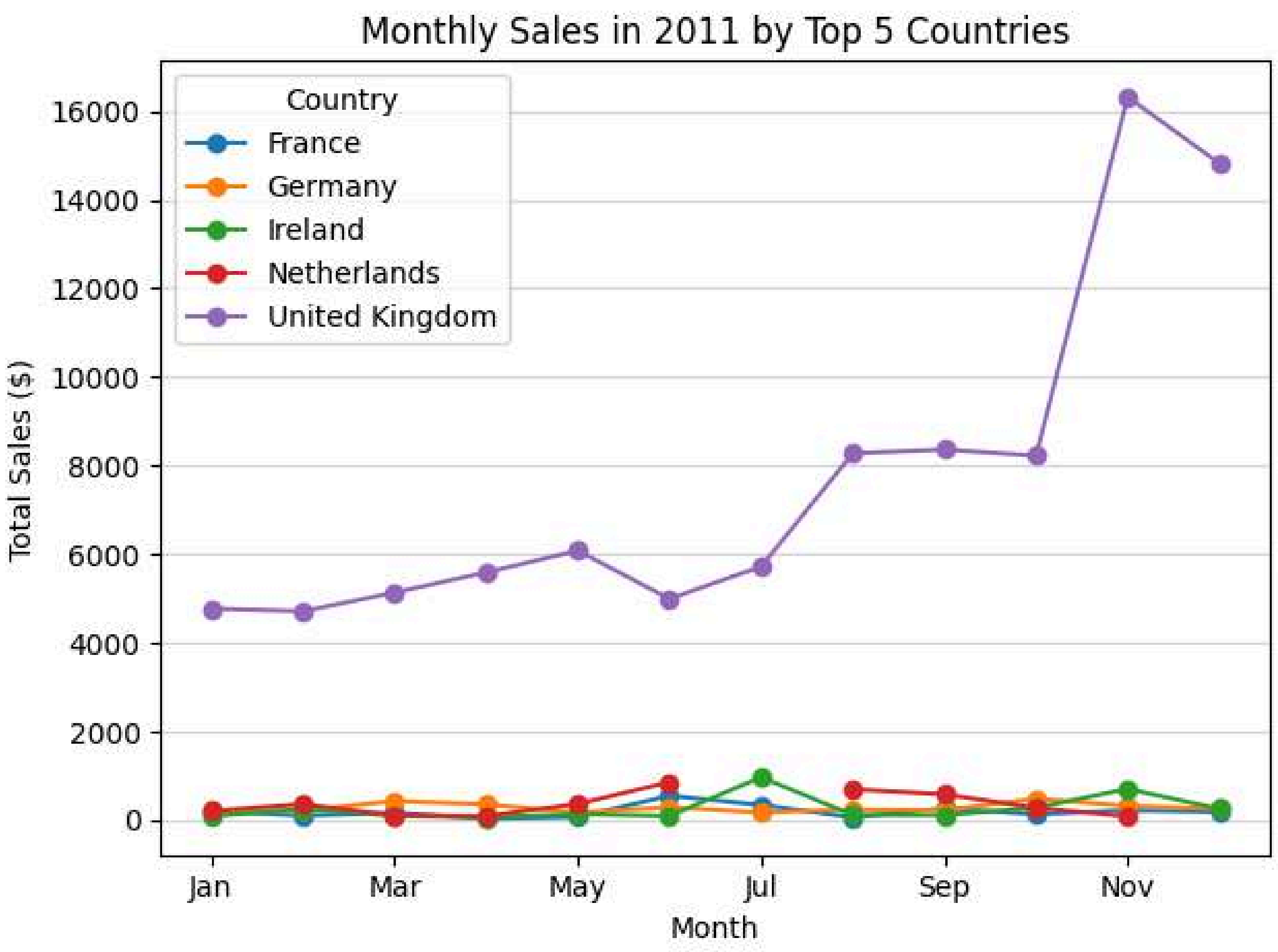
1. The shipping cost is way cheaper in the UK compared to other nations;
2. Low awareness of e-commerce's existence in other nations compared to the UK;
3. International customers prefer purchasing gifts from physical stores rather than from an online store where one has to wait for days/weeks just for the items to be delivered.



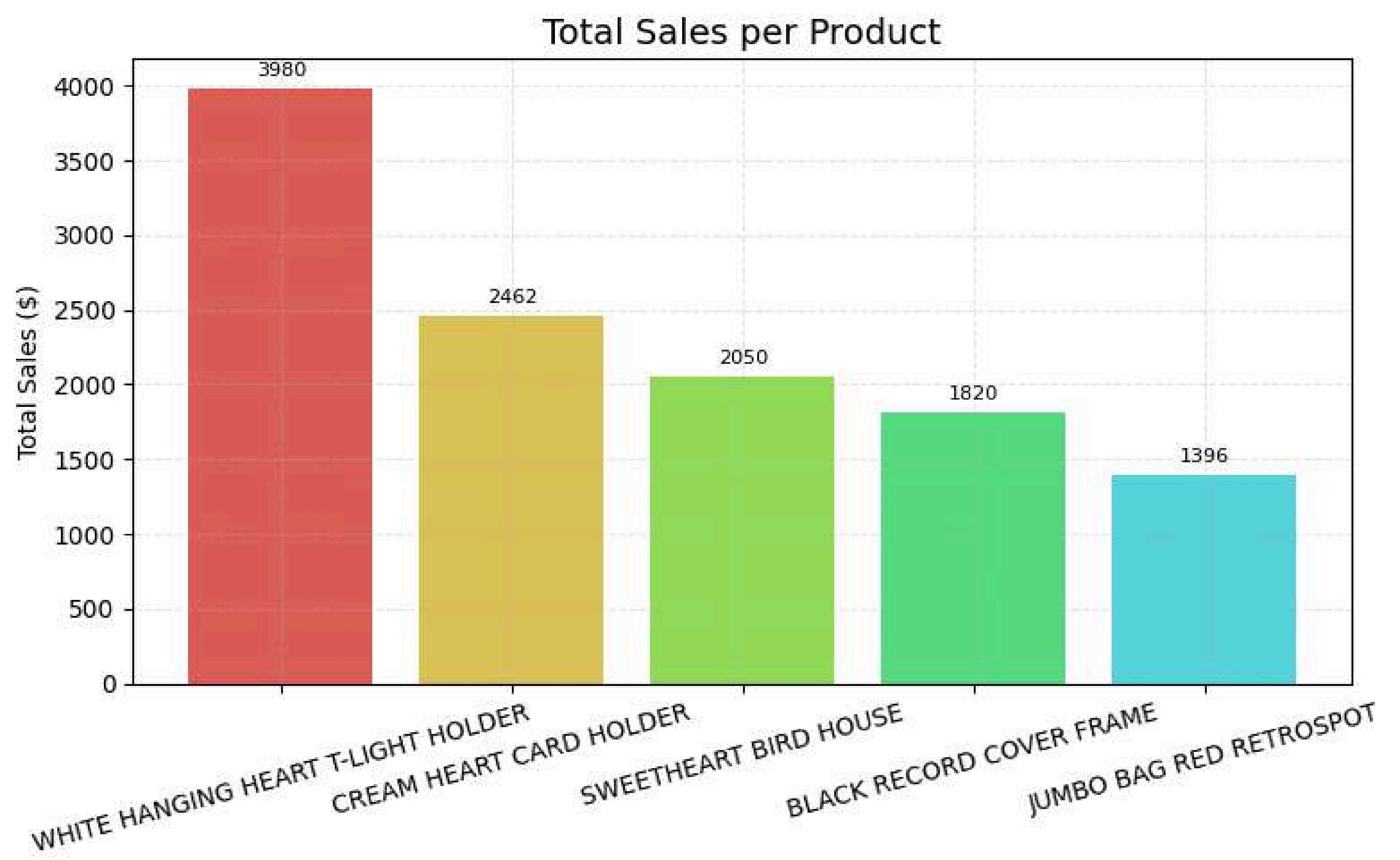
Monthly Sales by Top 5 Countries

Among the top 5 countries by sales, the UK shows robust performance, peaking in November and falling in June. The Netherlands has no sales in July or December, in contrast to Ireland, which performs best in July. On the other hand, Germany's and France's sales are mostly stable throughout the months.

In the Netherlands, people tend to spend time offline with their families, friends, and loved ones during holidays. This is reflected in the government's policy, where no business open except offline retail during summer holiday in July and the winter holiday in December; therefore, the company made no revenue in those respective months.

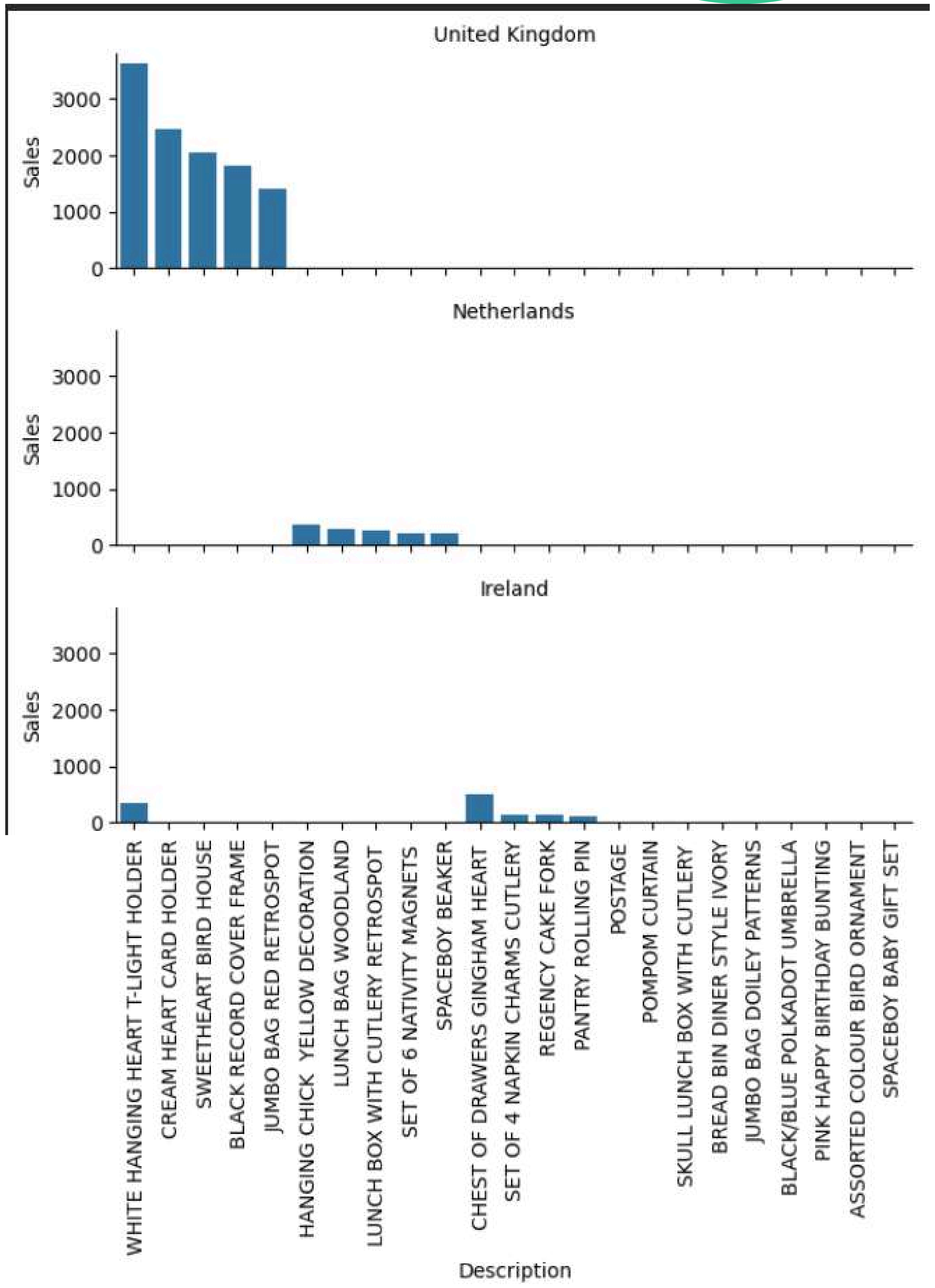


Overall Best-Selling Products



Among the 5 best-selling products, the White Hanging Heart T-Light Holder is the most popular. It made 3,979.65 USD in sales and 1,291 purchase orders.

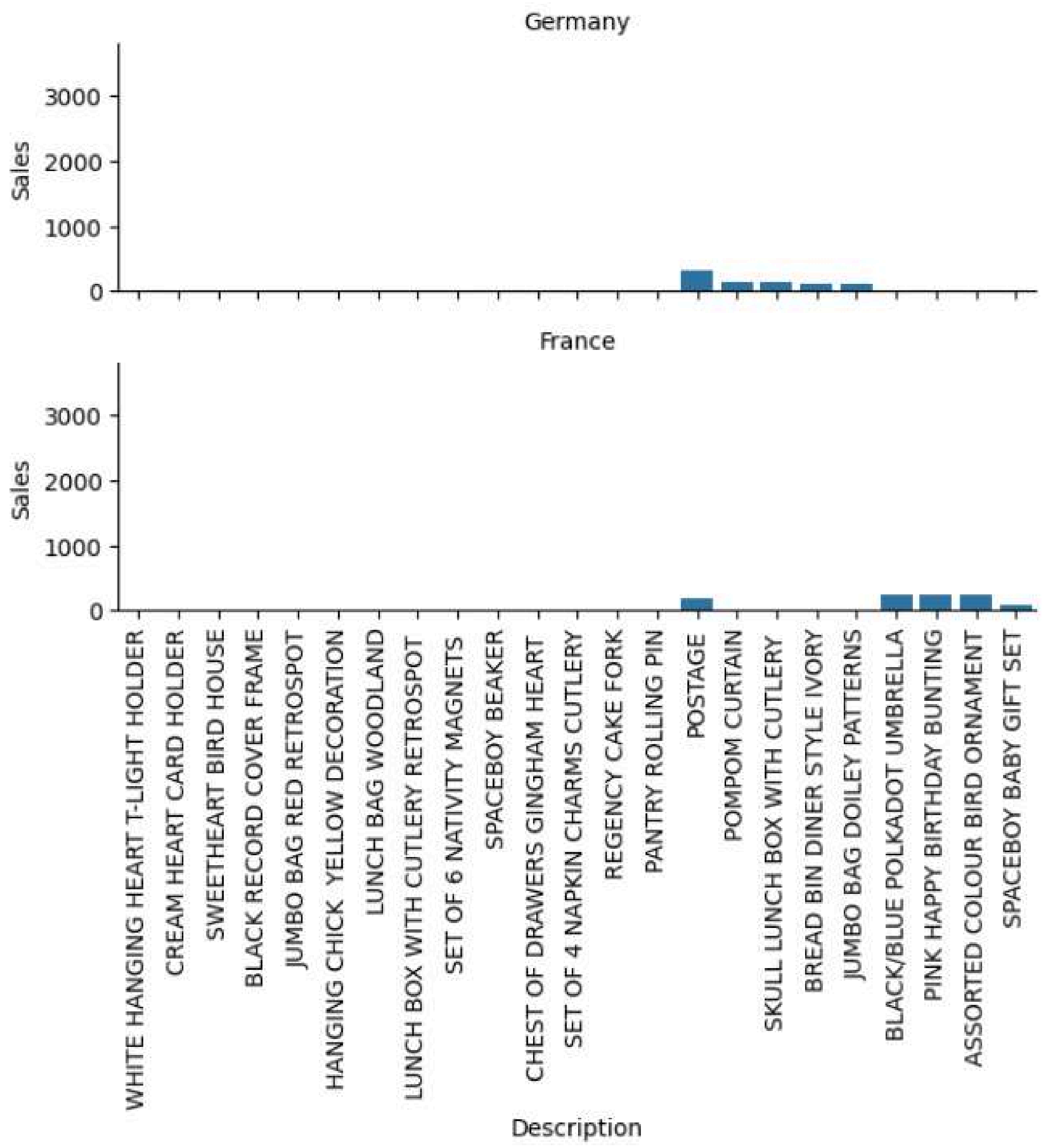
This T-Light Holder is a unique and thoughtful decoration for special occasions such as parties, celebrations, romantic dinners, or simply as a room accessory. Since the quantity is significantly higher than that of the other items, there are some indications of bulk purchases of T-Light Holder products.



Top 5 Products per Top 5 Countries

Each regions have their own preference when it comes to popular products, yet the White Hanging Heart T-Light Holder is popular among British and Irish customers, with the UK as the more dominant market.
Both countries share a tea culture where tea light usually purchased for this kind of occasion.

Top 5 Products per Top 5 Countries



Postage products generated more sales in both Germany and France. This indicates that postal collections and snail mail are still popular among people in the respective countries.

Business Recommendations

1. Increase product stock and warehouse capacity support during peak seasons (e.g., summer holidays, Halloween, Christmas) to prevent understocking and warehouse overload. Push the marketing campaigns during this time to boost sales.

2. Put the best-selling products on the homepage where web visitors can easily spot them. Customize the recommendations according to regional preferences.

3. Experiment with A/B testing for less popular items by offering voucher discounts and bundling options that contain both less popular products and popular products as the leverage.

4. Offer free shipping options with minimum orders to increase international sales.

5. Adjust the payment options and delivery options according to the region’s currency, payment systems, and local logistics services.

6. Examine the patterns that indicate wholesale purchasing and separate them from retail purchasing. Propose a B2B partnership to customers who frequently order in large bulk.

Limitations / Challenges

1. The data is only available in one year (2011). To uncover the patterns more accurately, it needs at least three consecutive years.

2. No customer ID column that indicates whether a transaction is made by a different person or a repeated order by the same person.

3. No unit column to distinguish the measurement unit, which can mislead the sales quantity of popular items over less popular items during the analysis process.

4. The analysis is based on the internal data only. Further information about the cause of the low sales trend (or even no sales at all, like what happened in the Netherlands) does not include external factors such as brand awareness, shipping costs, the government's trade policies, website traffic, ad ROI, etc.

Thank You

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